Conformal Prediction Under Distribution Shift

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Abstract

Split-conformal prediction gives regression models finite-sample, distribution-free prediction intervals with a guaranteed coverage rate, but the guarantee relies on the calibration and test data being exchangeable. This project asks a direct question: *what happens to that coverage guarantee when the test distribution no longer matches the calibration distribution?* Using synthetic nonlinear regression data with a known ground-truth function, we build 90% split-conformal intervals, confirm they behave correctly with no shift, and then deliberately introduce three kinds of distribution shift – covariate shift, noise shift, and subgroup (mixture) shift – to measure how badly empirical coverage degrades. We then test four candidate fixes: recalibrating on the shifted distribution, calibrating per subgroup, using a larger calibration set, and changing the base regression model. Recalibration and group-specific calibration both restore coverage close to nominal, but recalibration does so by inflating interval width dramatically, exposing a coverage-vs-usefulness trade-off that is easy to miss if only coverage is reported.

1 Introduction

Point predictions from a regression model say nothing about how much to trust them. Conformal prediction addresses this by wrapping any trained model with a procedure that produces prediction intervals carrying a formal, finite-sample coverage guarantee: under the sole assumption that calibration and test data are exchangeable (informally, drawn from the same distribution, in any order), a $(1-\alpha)$ split-conformal interval covers the true outcome at least $(1-\alpha)$ of the time, regardless of the base model or the data-generating process.

That guarantee is powerful but fragile: it depends entirely on exchangeability. In real deployments, the data a model sees in production routinely drifts away from the data it was calibrated on – new user populations, sensor drift, seasonal effects, dataset shift between collection sites. When that happens, nothing in the conformal machinery prevents coverage from silently collapsing. The goal of this project is to make that failure mode visible, quantify it under controlled synthetic conditions, and evaluate whether simple, practical fixes can restore it.

2 Background on Conformal Prediction

Split-conformal prediction (Papadopoulos et al., 2002; Lei et al., 2018) works as follows for regression:

1. Fit a base model $\hat{f}$ on a training set.
2. On a separate calibration set of size n, compute the nonconformity scores $r_i = |y_i - \hat{f}(x_i)|$, i.e. the absolute residuals.

3. Take the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the r_i's, call it $\hat{q}$. This finite-sample correction (rather than the plain $(1-\alpha)$ quantile) is what makes the guarantee exact rather than asymptotic.

4. For a new point x, output the interval $[\hat{f}(x) - \hat{q},\ \hat{f}(x) + \hat{q}]$.

Coverage guarantee. If the calibration points and the test point are exchangeable, then

$$\Pr\left(y_{\text{test}} \in [\hat{f}(x_{\text{test}}) - \hat{q},\ \hat{f}(x_{\text{test}}) + \hat{q}]\right) \geq 1 - \alpha.$$

The guarantee is *marginal* (averaged over the randomness in calibration and test draws), *distribution-free* (no assumption on $\hat{f}$ or on the shape of the residual distribution), and holds in finite samples, not just asymptotically.

Where it breaks. Exchangeability is exactly the assumption that fails under distribution shift: if the test distribution differs from the calibration distribution, the calibration residuals r_i are no longer representative of the residuals the model will actually make at test time, so $\hat{q}$ is calibrated to the wrong error scale. The theorem simply no longer applies – it does not degrade gracefully by itself. This project measures how large that gap becomes as a function of shift severity, and evaluates fixes that attempt to restore some form of the guarantee.

3 Method

All code is organized into small, single-purpose modules: `data_generation.py` (synthetic data and shift generators), `conformal.py` (calibration and evaluation), `experiments.py` (every experiment below), `plots.py` (figures), and `Main.py` (orchestration, which reproduces every number and figure in this report end to end).

The overall pipeline is:

1. Generate synthetic (X, y) data from a known nonlinear function (Section 4).

2. Split into training, calibration, and test sets.

3. Train a base regression model on the training set.

4. Build a 90% split-conformal interval width from the calibration set (Section 2).

5. Evaluate empirical coverage and average interval width on test data drawn from the *same* distribution as calibration (Section 5, the baseline), and then on test data drawn from increasingly *shifted* distributions (Section 6).

6. Attempt four fixes and measure whether each restores coverage, and at what cost in interval width (Section 7).

Throughout, the target coverage is $1 - \alpha = 90\%$ ($\alpha = 0.10$), and every reported "coverage" is the empirical fraction of test points whose true y falls inside its predicted interval.

4 Synthetic Data Setup

Ground truth is known because the data is generated by us. Each input $x = (x_0, x_1, x_2, x_3)$ is drawn i.i.d. from $\mathcal{N}(x_{\text{loc}}, x_{\text{scale}}^2)$, and the target is

$$y = 2x_0 - 1.5x_1 + 0.8x_2^2 + \sin(2x_3) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \text{noise\_std}^2).$$

The quadratic and sinusoidal terms make the relationship nonlinear, so a base model has to actually learn structure rather than fit a straight line; a random forest (300 trees) is used as the default base model except in the model-comparison experiment (Section 7.4), where linear regression and gradient boosting are added.

Baseline (unshifted) settings are $x_{\text{loc}} = 0$, $x_{\text{scale}} = 1$, noise_std = 1. Data set sizes are 1500 training points, 500 calibration points, and 1000 test points, unless an experiment explicitly varies one of these (e.g. Section 7.3).

Three kinds of shift are used, each isolating a different real-world failure mode:

- **Covariate shift**: increase x_{loc} from 0 to 3, moving test inputs into a region the model saw little or no training data for.
- **Noise shift**: increase noise_std from 1 to 3, keeping inputs in-distribution but making the labels themselves noisier and thus intrinsically harder to predict.
- **Subgroup (mixture) shift**: the population is a mixture of a low-noise group A (noise_std = 1.0) and a high-noise group B (noise_std = 2.5); the *mixture weight* of group B is increased from 20% to 80% between calibration time and test time, while each group's own conditional distribution never changes.

5 Baseline Experiment

With no shift, the random forest is calibrated at the nominal 90% level and evaluated on a fresh unshifted test set:

Quantity	Value
Target coverage	90.0%
Conformal half-width $\hat{q}$	2.096
Empirical coverage	92.2%
Average interval width	4.192

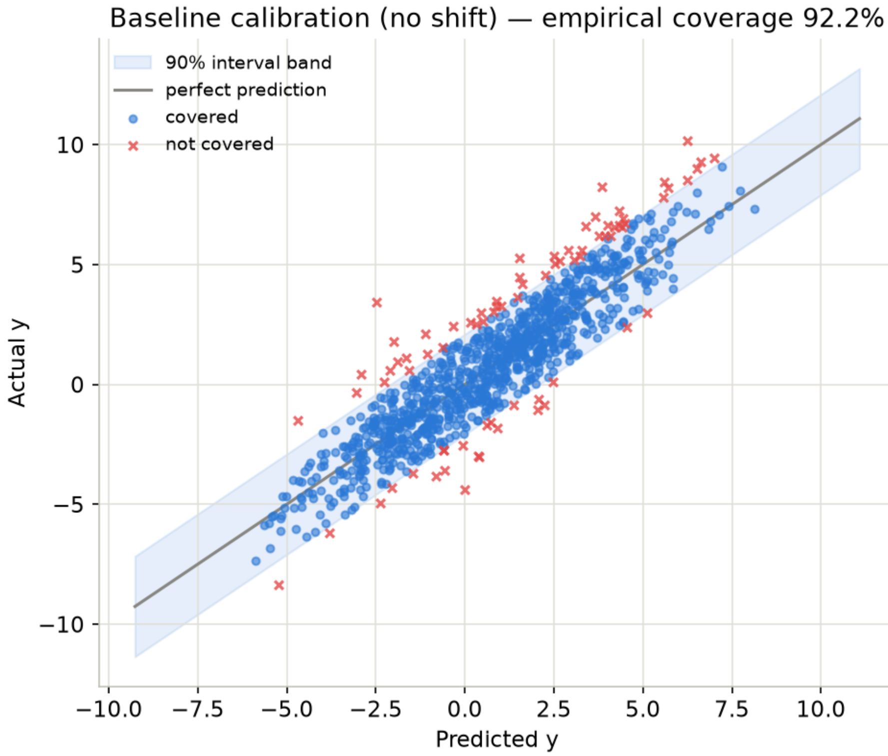

Figure 1: Predicted vs. actual y on the unshifted test set. The shaded band is the $\pm\hat{q}$ conformal band around perfect prediction; blue points fall inside their interval, red crosses do not. Empirical coverage (92.2%) is at or slightly above the 90% target, exactly as the theory predicts under exchangeability.

Coverage lands a bit above the nominal target, which is expected: the finite-sample correction $\lceil (n+1)(1-\alpha) \rceil / n$ rounds *up*, so split-conformal intervals are guaranteed to hit *at least* $1-\alpha$ coverage in expectation, not exactly $1-\alpha$.

6 Distribution-Shift Experiments

6.1 Covariate shift

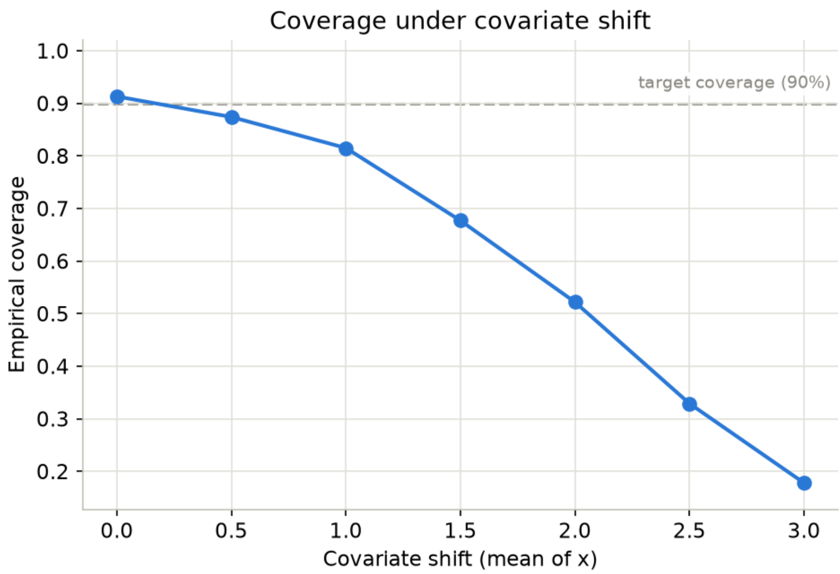

Figure 2: Empirical coverage as the test input mean x_{loc} moves from 0 to 3, using the interval width calibrated at $x_{\mathrm{loc}} = 0$.

Shift (x_{loc})	Coverage	Avg. width
0.0	91.3%	4.192
1.0	81.5%	4.192
2.0	52.2%	4.192
3.0	17.8%	4.192

Coverage collapses from above target to below one-fifth as inputs move away from the training region. The interval width never changes because it is fixed at calibration time – the model's predictions simply become increasingly biased on inputs it has not seen, and a fixed-width band around a biased prediction misses the truth far more often.

6.2 Noise shift

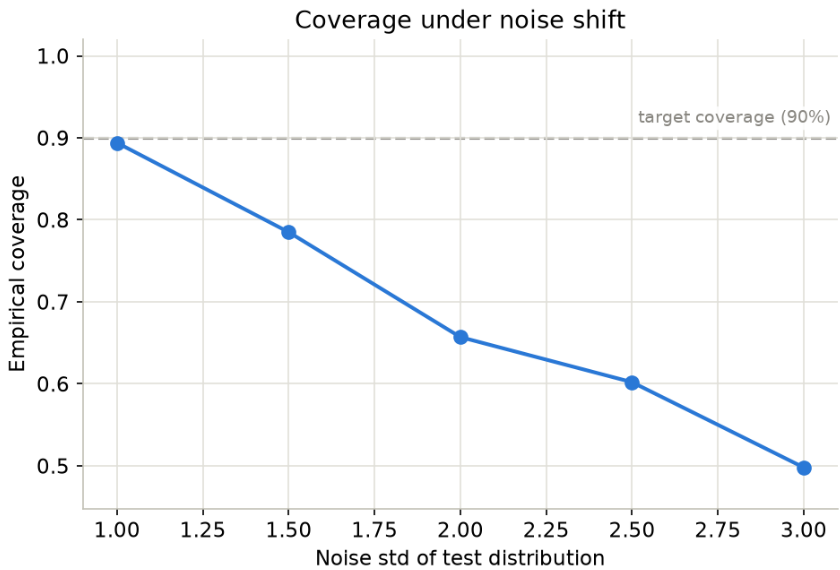

Figure 3: Empirical coverage as the test-label noise std grows from 1 to 3, with inputs kept in-distribution.

Noise std	Coverage	Avg. width
1.0	89.4%	4.192
2.0	65.7%	4.192
3.0	49.8%	4.192

Even with inputs fully in-distribution, coverage still fails once the intrinsic label noise at test time exceeds what calibration prepared for – the calibrated width simply is not wide enough for the new noise level.

6.3 Subgroup shift

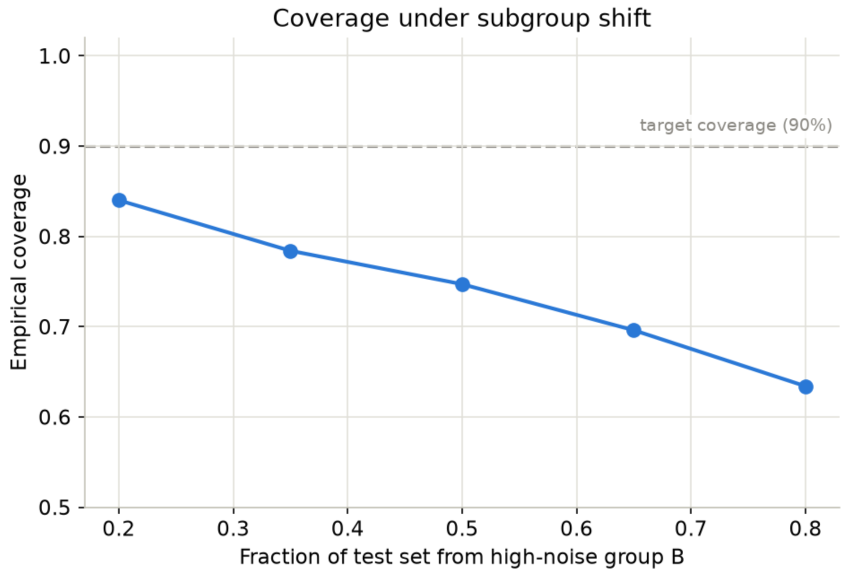

Figure 4: Empirical coverage as the fraction of high-noise group B in the test population grows, using a width calibrated on a homogeneous (single-noise-level) distribution.

Fraction group B	Coverage	Avg. width
0.20	84.0%	4.192
0.50	74.7%	4.192
0.80	63.4%	4.192

Here the calibration set contains no group-B-like (high-noise) points at all, so even a 20% contamination by group B already pulls coverage below target; as that fraction grows toward 80%, coverage keeps falling. This experiment is distinct from the group-specific fix in Section 7.2, which instead calibrates on a *known* A/B mixture and studies what happens when that mixture proportion shifts – a subtly different and, as it turns out, much more repairable problem.

7 Attempted Fixes

7.1 Recalibration

If we are willing to draw a fresh calibration set from the *shifted* distribution itself and recompute $\hat{q}$, does coverage recover?

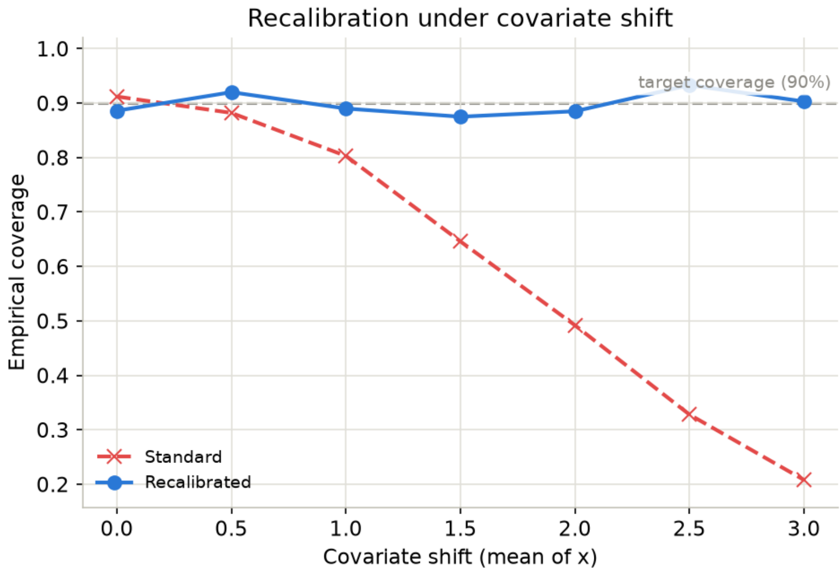

Figure 5: Standard (fixed calibration) vs. recalibrated coverage under covariate shift.

Shift	Cov. (standard)	Width (standard)	Cov. (recalibrated)	Width (recalibrated)
0.0	91.2%	4.192	88.6%	3.838
1.0	80.3%	4.192	89.0%	5.788
2.0	49.2%	4.192	88.5%	13.128
3.0	20.8%	4.192	90.3%	27.993

Yes – recalibrated coverage stays close to 90% at every shift level. But the average interval width grows from 3.84 to **28.0**, nearly a sevenfold increase. Recalibration does not make the model better at the shifted task; it simply admits, correctly, that the model's predictions are far less reliable there, and widens the interval enough to compensate. Reporting coverage alone would make this look like a clean fix; reporting width alongside it reveals that the "fix" is really just honest uncertainty quantification of a model operating far outside its competence.

7.2 Group-specific calibration

For the subgroup-mixture setting, we instead calibrate once on a known baseline mixture (80% group A, 20% group B), computing both a pooled width and a per-group width (A: 1.981, B: 4.753, vs. pooled: 2.545), and then evaluate both as the test-set mixture shifts toward group B.

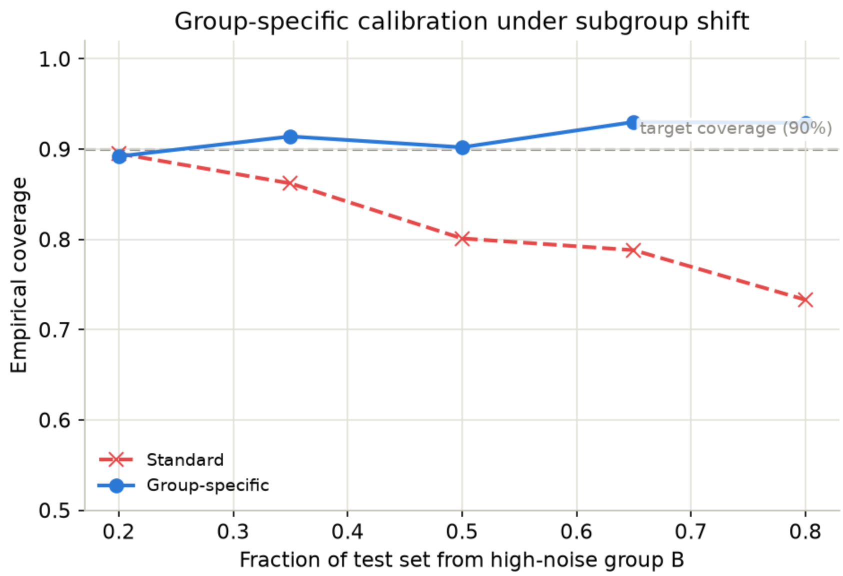

Figure 6: Standard (pooled) vs. group-specific coverage as the test population's group-B fraction grows from its calibration-time value (20%) to 80%.

Frac. group B	Cov. (pooled)	Width (pooled)	Cov. (group-specific)	Width (group-specific)
0.20	89.5%	5.091	89.2%	5.071
0.50	80.1%	5.091	90.2%	6.734
0.80	73.3%	5.091	92.9%	8.397

At the calibration-time mixture (20%), pooled and group-specific calibration agree, as expected. As the mixture shifts, pooled coverage drops to 73.3% because the single width was tuned to a mixture that no longer matches the test population. Group-specific calibration is essentially immune to this: because each test point is scored against *its own group's* width, and each group's own conditional noise level never changed, coverage stays at or above target regardless of the mixture proportion. This is the cleanest fix in the project, but it requires knowing group membership at both calibration and test time – see Section 9.

7.3 Calibration set size

Does using a larger calibration set (drawn from the shifted distribution, at covariate shift = 2.0) make recalibration more reliable? We repeat recalibration 20 times at each size and report the mean and standard deviation of achieved coverage.

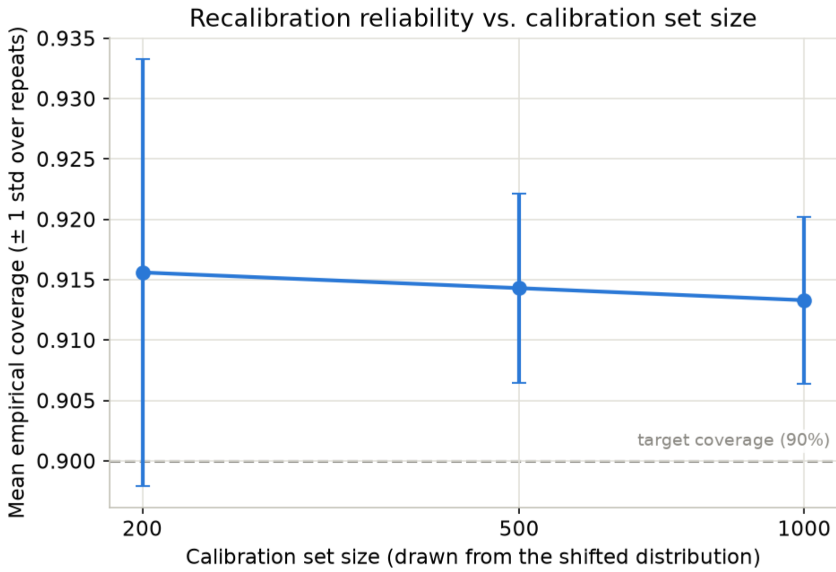

Figure 7: Mean achieved coverage (± 1 std over 20 repeats) after recalibrating with calibration sets of size 200, 500, and 1000 drawn from the shifted distribution.

Calib. size	Mean coverage	Std. coverage	Mean width	Std. width
200	91.6%	1.8pp	14.04	1.36
500	91.4%	0.8pp	13.94	0.68
1000	91.3%	0.7pp	13.86	0.61

The *mean* achieved coverage barely moves with calibration size – even 200 points is enough to recalibrate correctly on average. What changes is *reliability*: the standard deviation of achieved coverage across repeated draws shrinks from 1.8 percentage points at $n = 200$ to 0.7 at $n = 1000$. In a real deployment you only get to draw one calibration set, so this variance is exactly the risk of an unlucky draw silently under- or over-covering; a larger calibration set is what buys protection against that, not a better point estimate.

7.4 Does the choice of base model matter?

Finally, we compare linear regression, random forest, and gradient boosting, all conformalized the same way, under noise shift.

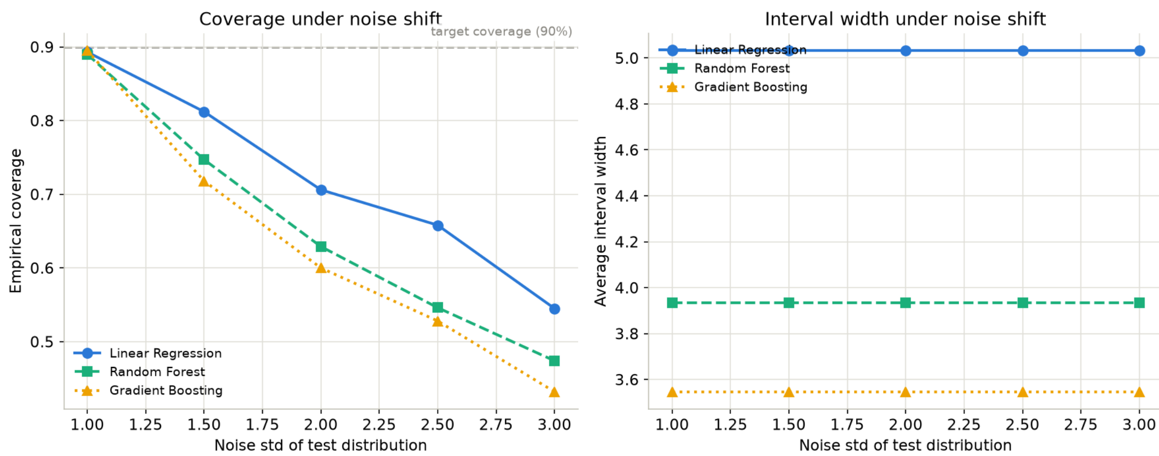

Figure 8: Left: coverage under noise shift for three base models. Right: their (constant, calibration-time) average interval widths.

Model	Coverage @ noise=1.0	Coverage @ noise=3.0	Width (constant)
Linear Regression	89.3%	54.5%	5.033
Random Forest	89.0%	47.4%	3.935
Gradient Boosting	89.5%	43.2%	3.547

All three models degrade in a similar pattern – coverage failure under shift is a property of the conformal procedure's exchangeability assumption, not of any particular model's flexibility. What does differ across models is the *in-distribution* interval width: gradient boosting's tighter in-sample residuals buy a noticeably narrower interval (3.55 vs. 5.03 for linear regression) for the *same* nominal coverage, but that narrower interval offers no extra robustness once the distribution shifts.

8 Results Summary

Two findings stand out across every experiment:

1. **Coverage is not free-standing – it must always be read next to interval width.** Both successful fixes (recalibration, group-specific calibration) restore coverage to nominal, but only by widening intervals, sometimes drastically (up to 7× under large covariate shift). A method that reports only coverage can look like it "solved" distribution shift while actually just returning uselessly wide intervals.

2. **Knowing the structure of the shift matters more than brute-force recalibration.** Group-specific calibration, which exploits known subgroup structure, restores coverage *without* the runaway width growth seen under generic recalibration, because it only widens intervals for the specific subpopulation that actually needs it.

9 Limitations

- All data is synthetic, from a single hand-chosen nonlinear function; real distribution shift is rarely this cleanly parameterized.

- Intervals are symmetric and fixed-width (standard split-conformal); no locally-adaptive methods (e.g. conformalized quantile regression, normalized/ studentized nonconformity scores) were tried, which would likely narrow intervals in easy regions even under shift.
- The recalibration fix assumes access to a *fresh, labeled* calibration set drawn from the shifted distribution at the moment of shift – an idealized, somewhat unrealistic assumption. Real deployments usually do not have labeled data from the new distribution immediately available.
- Group-specific calibration requires known group membership at both calibration and test time; this will not be available in most real settings.
- Most shift sweeps use a single random draw per shift level (only the calibration-size experiment repeats draws), so reported coverage values carry unquantified Monte Carlo noise on top of the systematic shift effect.
- Base models use largely default hyperparameters; no tuning was performed, so absolute width numbers (not the qualitative trends) would change with a more carefully tuned model.

10 Future Work

- Implement *weighted conformal prediction* (Tibshirani et al., 2019), which reweights calibration residuals by an estimated likelihood ratio between calibration and test covariate distributions, and does not require fresh labeled test-distribution data the way the recalibration fix here does.
- Implement *adaptive conformal inference* (Gibbs & Candès, 2021), which updates the interval width online from observed coverage as data streams in, for settings where shift is gradual rather than a single step change.
- Replace fixed-width intervals with conformalized quantile regression or normalized nonconformity scores, so interval width adapts to local difficulty rather than being constant across the input space.
- Repeat every shift sweep with multiple random seeds and report confidence bands, not single point estimates.
- Move from synthetic data to a real dataset with a natural train/test shift (e.g. a temporal or geographic split), to check whether these findings transfer.
- Explore subgroup-shift robustness when group labels are *not* known, e.g. by clustering residuals or covariates to discover latent subgroups before calibrating per cluster.